TokenSkip

Controllable Chain-of-Thought Compression in Large Language Models

Data Enthusiasts

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Problem: Inefficient Long Chain-of-Thought Reasoning

Chain-of-Thought (CoT) improves reasoning:

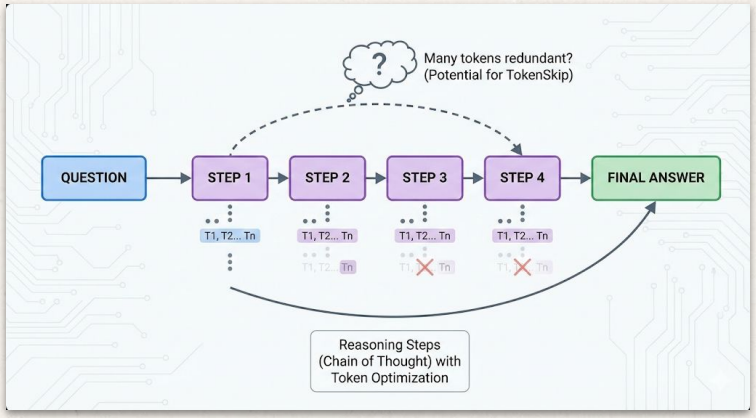
- Enables step-by-step problem solving
- Boosts accuracy in math & logic tasks
- Used in models like o1, DeepSeek-R1

However… Longer CoT creates problems

- Linear increase in inference latency
- Larger memory usage (KV cache growth)
- Higher computational cost (quadratic attention)

Core Problem:

- CoT sequences can exceed **thousands of tokens**
- Many tokens are **semantically redundant**
- Autoregressive decoding makes every token expensive



Central Research Question: Do all tokens in a Chain-of-Thought contribute equally to reasoning?

Longer reasoning improves performance — but at significant computational cost. Can we reduce CoT length without hurting accuracy?

Proposed Solution: TokenSkip

Core Idea:
Skip semantically unimportant reasoning tokens while preserving critical reasoning steps.

Step 1 — Token Importance Measurement

- Uses **LLMLingua-2**
- Assigns importance score to each CoT token
- Higher score → More critical to reasoning

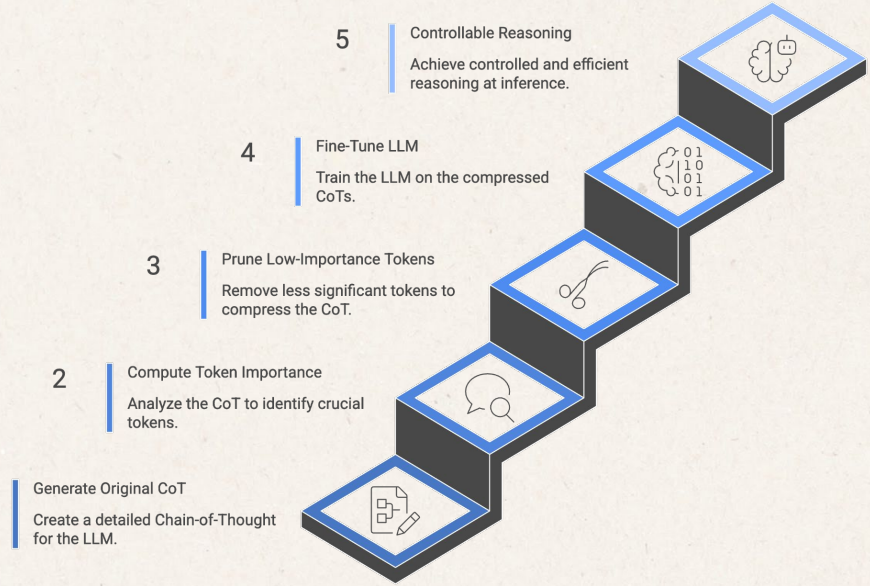
Step 2 — Token Pruning

- Select compression ratio $\gamma \in \{0.5–1.0\}$
- Keep top γ-quantile tokens
- Remove low-importance tokens

Result: Compressed CoT

Step 3 — Fine-Tuning for Control

- Fine-tune LLM on compressed CoTs
- Include γ in prompt
- Model learns to generate compressed reasoning natively
- Enables **controllable CoT compression**



Instead of truncating reasoning, TokenSkip learns shortcuts between critical reasoning tokens.

Experimental Setup

Models Used

- Base Model: *Llama-3.1:7B*
- Fine-tuning Method: LoRa
- Framework: HuggingFace Transformers
- Hardware: *GPU T4*2 provided by Kaggle*

Datasets

- MATH-500 (Advanced Math Reasoning)

Token Importance Metric

- LLMLingua-2 based scoring
- Token-level importance calculation
- Quantile-based pruning (γ compression ratio

Compression Settings

- Tested compression ratios: γ = {0.5, 0.6, 0.7, 0.8, 0.9, 1.0}
- Maintained answer tokens unchanged
- Used greedy decoding for inference

We trained the model on compressed reasoning traces to enable controllable CoT generation during inference.

MATH-500: Reproduced Results & Comparison with Paper

MATH-500 Results Summary (Our Implementation)

```
================================================================
MATH-500 RESULTS SUMMARY
================================================================
     Method Ratio Accuracy (%) Tokens Latency (s) ActRatio
   original     -         66.0 554.32       47.29        -
  BeConcise     -         72.0 393.38       33.62     0.71
OnlyNumbers     -         72.0 387.64       33.24     0.70
 AbbreWords     -         70.0 385.14       32.84     0.69
 Truncation     -         66.0 554.32       47.17        -
 Truncation   0.9         64.0 544.48       46.19     0.98
 Truncation   0.7         58.0 505.18       43.03     0.91
 Truncation   0.5         44.0 433.54       36.85     0.78
```

Comparison with Paper (Table 1 – MATH-500)

```
================================================================
COMPARISON WITH PAPER RESULTS (Table 1, MATH-500)
================================================================
Method          Paper Acc    Our Acc      Diff
-------------------------------------------------
original        48.6         66.0         +17.4 ~
BeConcise       47.4         72.0         +24.6 ~
OnlyNumbers     46.4         72.0         +25.6 ~
AbbreWords      47.6         70.0         +22.4 ~

Note: Small differences are expected due to sampling and hardware variations.
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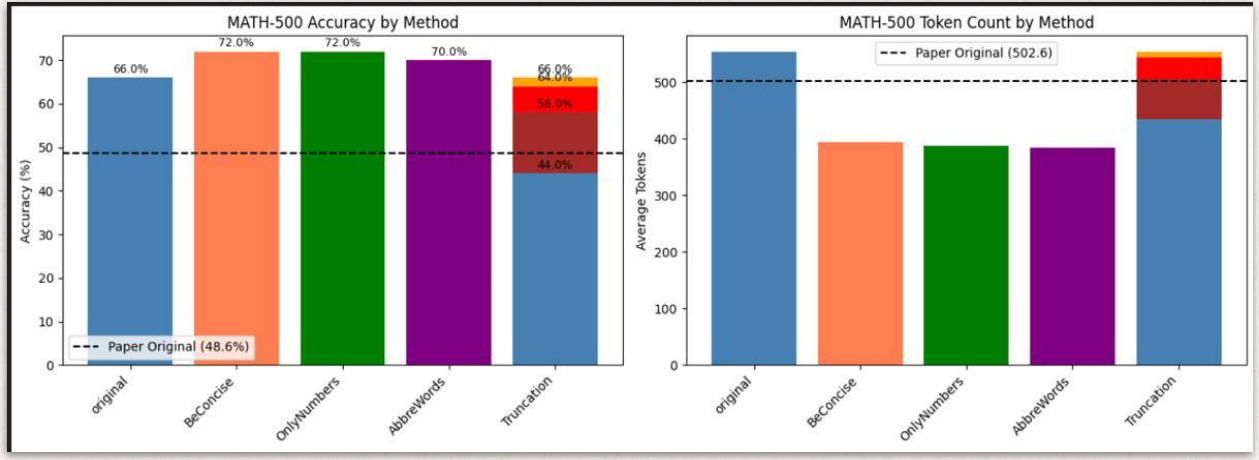

Key Insights:

- Prompt-based compression significantly improved accuracy in our setup
- Moderate token reduction (~30%) without accuracy loss
- Aggressive truncation leads to sharp performance drop
- Results differ from paper due to environment and model differences

Limitations / Gaps

- **Scale Constraints:** Due to computational limits, the authors did not evaluate TokenSkip on very large models such as Qwen2.5-72B or specialized long-reasoning models like QwQ-32B-Preview. This limits understanding of scalability on state-of-the-art reasoning systems.

- **Domain Sensitivity:** The token importance metric (LLMLingua-2) was not specifically trained on mathematical reasoning data. This may result in suboptimal pruning of numerical expressions and critical equations.

- **Adherence at Low Ratios:** Model performance and compression ratio adherence degrade significantly at extreme compression levels (below 0.5). This indicates instability when too many reasoning tokens are removed.

- **Training Requirement:** Unlike zero-shot prompting methods, TokenSkip requires supervised fine-tuning (though lightweight: ~2–2.5 hours). This introduces additional setup complexity and resource dependency.

- **Limited Domain Evaluation:** Experiments primarily focus on mathematical reasoning benchmarks (GSM8K, MATH-500). Generalization to domains such as code generation, multi-hop QA, or long-context reasoning remains underexplored.

Proposed Improvements / Extensions

- **Domain-Specific Importance Metrics:**
  - Train LLMLingua-2 on mathematical data
  - Use entropy-based scoring instead of generic BERT
  - Preserve numerical tokens and equations with higher priority
- **Scaling Law Analysis:**
  - Test TokenSkip on smaller models (1.5B, 3B) to find "collapse threshold"
  - Identify minimum model size for effective CoT compression
- **Financial Sentiment:**
  - Apply TokenSkip to financial reasoning (FiQA, FinancialPhraseBank)
  - Measure "Time-to-Signal" for trading applications
- **Adaptive Compression Ratio:**
  - Dynamic γ selection based on problem difficulty
  - Easy problems → aggressive compression (γ=0.5)
  - Hard problems → conservative compression (γ=0.9)
- **Syntax-Aware Pruning:**
  - Protect critical syntax tokens (brackets, operators, "NOT")
  - Add grammar constraints to importance scoring
  - Evaluate on code generation and logic puzzles
- **Long-CoT Model Evaluation:**
  - Test on QwQ-32B, DeepSeek-R1 with 10K+ token reasoning
  - TokenSkip may show greater benefits at extreme lengths

Based on continued research and experimentation, we will select and pursue one or more of these extensions to maximize both academic contribution and practical impact